

How to Read Linux Logs and Debug Issues Easily

Reading Linux logs is the most essential skill for troubleshooting server issues and app crashes. Once you learn a few basic terminal commands, finding root causes takes minutes instead of hours.

What you'll learn:

1. Watching Live Logs with Tail
2. Filtering Errors Fast with Grep
3. Inspecting Services with Journalctl
4. Navigating Large Files with Less

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1. Watching Live Logs with Tail

When an error occurs during runtime, opening a static text file won't show new updates. The tail command with the -f flag streams new log entries live to your terminal screen as your app writes them. Adding -n allows you to view the most recent historical lines before streaming starts.

```
$tail -f -n 50 /var/log/syslog
```

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2. Filtering Errors Fast with Grep

Log files usually contain thousands of normal operational lines that clutter your view. Using grep isolates specific keywords like ERROR, FAILED, or 500 to narrow down issues instantly. The -i flag ensures your search is case-insensitive so you never miss a capitalized keyword.

```
$grep -i "connection refused" /var/log/nginx/error.log
```

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3. Inspecting Services with Journalctl

Modern Linux distributions use systemd to manage background services and store logs centrally. Instead of hunting for text files in `/var/log`, you use `journalctl` to inspect service-specific activity. You can easily filter by service name, error severity, or specific time windows.

```
$journalctl -u nginx.service --since "1 hour ago" -p err
```

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4. Navigating Large Files with Less

Opening multi-gigabyte log files in text editors like nano or using cat can freeze your terminal session. The less command opens logs page-by-page without loading the whole file into system memory. You can navigate using arrow keys, search forward with /, and exit safely by pressing q.

```
$less /var/log/auth.log
```


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Key Takeaways

- + Pipe live tail outputs into grep to filter out normal non-critical log noise.
- + Check file read permissions with sudo if you encounter 'Permission denied' errors.
- + Use journalctl -u service_name first when debugging system services like Nginx or Docker.
- + Use case-insensitive search flags (-i) so you don't miss capitalized error keywords.
- + Redirect search outputs to a text file when sharing log snippets with senior engineers.

Watch Out For

- ! Running cat on massive log files, which overloads RAM and freezes the terminal.
- ! Forgetting that server log timestamps might be in UTC rather than your local timezone.
- ! Searching for vague terms like 'info' instead of precise keywords like 'exception' or 'failed'.

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